

AN APÉRY-LIKE SEQUENCE ATTACHED TO A SINGULAR $K3$ SURFACE

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ABSTRACT. The integer sequence $1, 4, 16, 64, 250, 928, 3136, 8704, 11866, -79400, \dots$ defined by the four-term recurrence $(n+1)^2 u_{n+1} = (11n^2 + 11n + 4)u_n - (37n^2 + 3)u_{n-1} + 3(3n-1)(3n-2)u_{n-2}$ is the period sequence of an elliptic $K3$ surface with five singular fibres I_4, I_6, I_3, I_3, IV^* and Picard number 20. Its transcendental motive is that of the weight-3 CM newform **32.3.d.a** (CM by $\mathbb{Q}(\sqrt{-2})$), and its Apéry constants are critical L -values of that form: the fold constants at the conjugate singularities $\frac{5 \pm i\sqrt{2}}{27}$ are $\frac{\sqrt{2} \pm i}{3} L(g, 2)$ and the constant at $t = 1$ is $\frac{2\sqrt{2}}{3} L(g, 2)$, with $L(g, 2) = \frac{\pi}{32} \Gamma(\frac{1}{8}) \Gamma(\frac{3}{8}) / \Gamma(\frac{5}{8}) \Gamma(\frac{7}{8})$ by Chowla–Selberg. To our knowledge this is the first Apéry-like sequence whose Apéry constant is a critical value of a cusp form; three-term integral recurrences can never produce one, by a Riemann–Hurwitz/twist-parity bound that confines them to rational elliptic surfaces.

1. THE SEQUENCE AND THE RESULTS

Let $u_{-1} = u_{-2} = 0$, $u_0 = 1$ and

$$(1) \quad (n+1)^2 u_{n+1} = (11n^2 + 11n + 4) u_n - (37n^2 + 3) u_{n-1} + 3(3n-1)(3n-2) u_{n-2}.$$

All u_n are integers (verified for $n \leq 300$; no proof is known — see Remark 3.1). The sequence arose in a systematic scan [4] of four-term Apéry-like recurrences: among 2178 integral rows found in the Kodaira-admissible classes, 98% are three-term rows in disguise (an apparent or I_0^* singular point) and, of the 30 genuinely five-point sporadic rows, (1) is the unique one whose local system is the Picard–Fuchs system of an elliptic surface.

Write $A(t) = \sum_{n \geq 0} u_n t^n$; it satisfies $LA = 0$ for the self-adjoint operator

$$Ly = (tR(t)y')' - 4(1 - 10t + 15t^2)y, \quad R(t) = 1 - 11t + 37t^2 - 27t^3 = (1-t)(1-10t+27t^2),$$

with singular points $t = 0, \frac{5 \pm i\sqrt{2}}{27}, 1, \infty$, all local exponents $(0, 0)$ at the finite points. Let $B(t) = \sum b_n t^n$ be the companion solution of $LB = 1$ with $b_0 = 0, b_1 = 1$; then $d_n^2 b_n \in \mathbb{Z}$ with $d_n = \text{lcm}(1, \dots, n)$, sharp.

Theorem 1.1. (i) *The local system of L is $R^1 \pi_* \mathbb{Q}$ of an elliptic $K3$ surface $\pi : \mathcal{X} \rightarrow \mathbb{P}_t^1$ with singular fibres $I_4(t=0)$, $I_6(t=1)$, $I_3, I_3(t = \frac{5 \pm i\sqrt{2}}{27})$, $IV^*(t=\infty)$, given by the explicit Weierstrass model $y^2 = x^3 - 27c_4x - 54c_6$ of §2. The functional invariant $\mathcal{J}$ is a Belyi map of degree 16 attaining the Riemann–Hurwitz bound.*

(ii) *$\rho(\mathcal{X}) = 20$ (a singular $K3$), MW has rank 0 and torsion exactly $\mathbb{Z}/3$, generated by a section over $\mathbb{Q}(\sqrt{3})$ with $x_0 = 9(4t-1)^2$; the transcendental lattice is $T \cong \text{diag}(6, 12)$, $\det T = 72$, with CM by $\mathbb{Q}(\sqrt{-2})$.*

(iii) *The transcendental motive is that of the newform $g = \mathbf{32.3.d.a} = \mathbf{8.3.d.a} \otimes \chi_{-4} \in S_3(\Gamma_0(32), \chi_{-8})$, $\mathbf{8.3.d.a} = \eta(z)^2 \eta(2z) \eta(4z) \eta(8z)^2$: the traces $a_p = \sum_{t, x \in \mathbb{F}_p} \chi_p(x^3 - 27c_4(t)x - 54c_6(t))$ agree with $a_p(g)$ for all 76 primes $5 \leq p \leq 397$ (Livné rigidity then identifies the motive).*

(iv) The fold constants of the pair (A, B) (defined in §3; verified to 165–230 digits) are

$$\xi\left(\frac{5 \mp i\sqrt{2}}{27}\right) = \frac{\sqrt{2} \mp i}{3} L(g, 2), \quad \xi(1) = \frac{2\sqrt{2}}{3} L(g, 2) = \xi_+ + \xi_-, \quad L(g, 2) = \frac{\pi}{32} \frac{\Gamma(\frac{1}{8})\Gamma(\frac{3}{8})}{\Gamma(\frac{5}{8})\Gamma(\frac{7}{8})}.$$

(v) 2-adically $b_n/a_n \rightarrow 0$ at logarithmic rate $v_2(b_n/a_n) = \log_2 n + O(1)$; there is no 3-adic limit.

Items (i), (ii) are proved; (iii) is proved modulo the standard application of Livné’s theorem, with the finite verification stated; (iv) is a numerical identification at the stated precision, cross-validated by two independent methods; (v) is measured. Full details, scripts and logs: [5, 4].

Remark 1.2. Every previously known Apéry-like sequence with identified Apéry constant computes a period of an *Eisenstein* class — $\zeta(2), \zeta(3)$, Dirichlet $L(2, \chi)$, $L(3, \chi)$ — or, for the square-root rows, an L -value of a weight-3 CM form arising on a *rational* elliptic surface host [1, 9]. Theorem 1.1 gives the first Apéry constant that is a critical cusp-form L -value attached to a $K3$. This is forced by the shape of the recurrence: for three-term rows the functional invariant satisfies $\deg \mathcal{J} \leq 12$ and a twist-parity argument confines the untwisted surface to Beauville’s rational list [3]; five singular points allow $\deg \mathcal{J} \leq 18$, and (1) realises a $K3$ with $\deg \mathcal{J} = 16$.

2. THE SURFACE

The scan’s $\mathcal{J}$ -map certification gives $\mathcal{J} = U/V$ with

$$V = t^4(t-1)^6(27t^2 - 10t + 1)^3, \quad U = -(4t-1)^3(216t^4 + 64t^3 - 48t^2 + 12t - 1)^3,$$

$U - 1728V = -(5832t^8 + 34560t^7 - 30016t^6 + 12624t^5 - 4380t^4 + 1280t^3 - 240t^2 + 24t - 1)^2$, fitted from 38 coefficients of the canonical nome and verified on 73 more ($h = 4$, $\gamma = -1$: $\mathcal{J}(t) = j(-q(t)^4)$). Branch reconstruction (with the quadratic twist pinned by the exact series identities $c_4 = E_4(Q)/y_0^4$, $c_6 = E_6(Q)/y_0^6$ at $Q = -q(t)^4$) yields

$$c_4 = (4t-1)(216t^4 + 64t^3 - 48t^2 + 12t - 1), \quad c_6 = -(5832t^8 + \cdots - 1), \quad \Delta = -V,$$

and Tate’s algorithm reads off the fibre table of Theorem 1.1(i); $\sum e = 24$, so $\mathcal{X}$ is $K3$. Shioda–Tate gives $\rho \geq 2 + 3 + 5 + 2 + 2 + 6 = 20 = h^{1,1}$, hence $\rho = 20$ and MW rank 0. The 3-division polynomial factors as $\psi_3 = 3(x - 9(4t-1)^2) \cdot (\text{irreducible cubic})$ with $y_0^2 = 432(t-1)^4(27t^2 - 10t + 1)^2$: a 3-torsion section over $\mathbb{Q}(\sqrt{3})$. The determinant count $\det T = 648/m^2$ with the even-lattice condition $\det T \equiv 0, 3 \pmod{4}$ and Shioda’s injection $\text{MW}_{\text{tors}} \hookrightarrow \prod_v G_v$ pin $m = 3$, $\det T = 72$; the discriminant-form computation in [5] identifies $T \cong \text{diag}(6, 12) = 3\langle x^2 + 2y^2 \rangle$, whence the CM field $\mathbb{Q}(\sqrt{-2})$.

For the trace identity in Theorem 1.1(iii): with $\mathcal{F} = R^1\pi_*\mathbb{Q}_\ell$ on the good locus, $H_c^0 = H_c^2 = 0$, and in $0 \rightarrow \oplus_v \mathcal{F}^{I_v} \rightarrow H_c^1 \rightarrow H^1(\mathbb{P}^1, j_*\mathcal{F}) \rightarrow 0$ the skyscraper contribution $\sum a_v$ cancels against the same terms in $-\sum_{t \in \mathbb{P}^1(\mathbb{F}_p)} a_t$, leaving $\text{tr}(\text{Frob}_p | T \otimes \mathbb{Q}_\ell)$ on the nose (here $H^1(\mathbb{P}^1, j_*\mathcal{F}) = T \otimes \mathbb{Q}_\ell$ because $\rho = 20$ and the Néron–Severi classes are accounted). The bad points remain distinct mod every $p \geq 5$ (the relevant resultants are supported on $\{2, 3\}$; $\text{disc}(27t^2 - 10t + 1) = -8$). The match with 32.3.d.a holds at all 76 primes $5 \leq p \leq 397$ with no free parameter, and $a_p = 0$ exactly at $p \equiv 5, 7 \pmod{8}$, as CM by $\mathbb{Q}(\sqrt{-2})$ requires.

3. THE PERIODS

Each finite singular point has exponents $(0, 0)$: a logarithmic fold. At such a point t_c write $A = c_0w_0 + c_1(w_0 \log(t - t_c) + h)$, $B = p + d_0w_0 + d_1(w_0 \log(t - t_c) + h)$, $\xi(t_c) := d_1/c_1$,

with w_0 the analytic Frobenius solution and p the analytic particular solution; $\xi(t_c)$ is independent of branch and normalisation. When t_c is the unique nearest singularity, $\xi(t_c) = \lim b_n/a_n$; here the nearest singularities are the conjugate pair $\frac{5 \pm i\sqrt{2}}{27}$ ($|t| = 27^{-1/2}$), so b_n/a_n oscillates and the ξ 's are complex fold constants. They were computed two independent ways — adaptive-step analytic continuation of the ODE with Frobenius matching at 230 digits, and directly from the recurrence asymptotics — and identified by integer-relation search against a basis containing $L(g, 2)$ and its twists, $\pi^2\sqrt{D}$, and the discriminant- (-8) Chowla–Selberg products; the identifications of Theorem 1.1(iv) hold with residual below 10^{-165} , and $L(g, 2)$ itself matches the Γ -quotient at 230 digits (Chowla–Selberg, as g has CM by $\mathbb{Q}(\sqrt{-2})$; cf. [2]). The relation $\xi(1) = \xi_+ + \xi_-$ is exact at the same precision. As a by-product the logarithm coefficients of A itself are algebraic multiples of $1/\pi$: $c_1(1) = -2/(\pi\sqrt{3})$, $c_1(t_+) = \frac{1}{\pi}(-\frac{7}{2\sqrt{3}} + \frac{4i}{\sqrt{6}})$.

Remark 3.1. The integrality of u_n is, at present, purely experimental ($n \leq 300$; also mod small primes to $n = 10^4$ in the scan). For three-term rows integrality is explained by modular parametrisation (the Hauptmodul is a congruence-group uniformiser); here the monodromy group is an index-16 genus-0 subgroup attaining the Riemann–Hurwitz bound (signature data in [5]), and we do not know whether it is congruence, nor a modular proof of integrality. This seems to us the most interesting open question the sequence raises: it is precisely Zagier's integrality phenomenon [9] one step beyond the rigid (three-term) case, where the classification of [3] no longer forces modularity.

Remark 3.2. 2-adically $b_n/a_n \rightarrow 0$: the cuspidal clause of the Euler-factor criterion of [7] ($\xi_p = 0$ for cuspidal classes), observed here for the first time on a cusp-form row. The rate is logarithmic, not linear: the slope law for Eisenstein rows (Proposition C of [7]) does not extend to cuspidal rows.

Reproducibility. All computations are in the repository `math-modular-sources`: the scan and the row's discovery in `consolidation/FOUR_TERM_SCAN.md` (`lattice/four_term/`), the surface, newform and period work in `consolidation/K3_ROW_PERIOD.md` (`lattice/k3_period/`), with PARI/GP and mpmath scripts and logs. The newform is 32.3.d.a in the LMFDB.

REFERENCES

- [1] F. Beukers, *Irrationality proofs using modular forms*, Astérisque 147–148 (1987), 271–283.
- [2] S. Chowla, A. Selberg, *On Epstein's zeta-function*, J. reine angew. Math. 227 (1967), 86–110.
- [3] *Integral second-order Apéry-like recurrences and rational elliptic surfaces*, companion paper (`paper/classification`), 2026.
- [4] *Four-term Apéry-like rows: the five-singular-point scan*, `consolidation/FOUR_TERM_SCAN.md`, 2026.
- [5] *The period of the elliptic- $K3$ four-term row*, `consolidation/K3_ROW_PERIOD.md`, 2026.
- [6] R. Livné, *Motivic orthogonal two-dimensional representations of $\text{Gal}(\overline{\mathbb{Q}}/\mathbb{Q})$* , Israel J. Math. 92 (1995), 149–156.
- [7] *Modular sources of Apéry-like sequences* (main project paper), `consolidation/paper_draft8.pdf`, 2026.
- [8] T. Shioda, H. Inose, *On singular $K3$ surfaces*, in: Complex Analysis and Algebraic Geometry, Cambridge (1977), 119–136.
- [9] D. Zagier, *Integral solutions of Apéry-like recurrence equations*, in: Groups and Symmetries, CRM Proc. 47 (2009), 349–366.